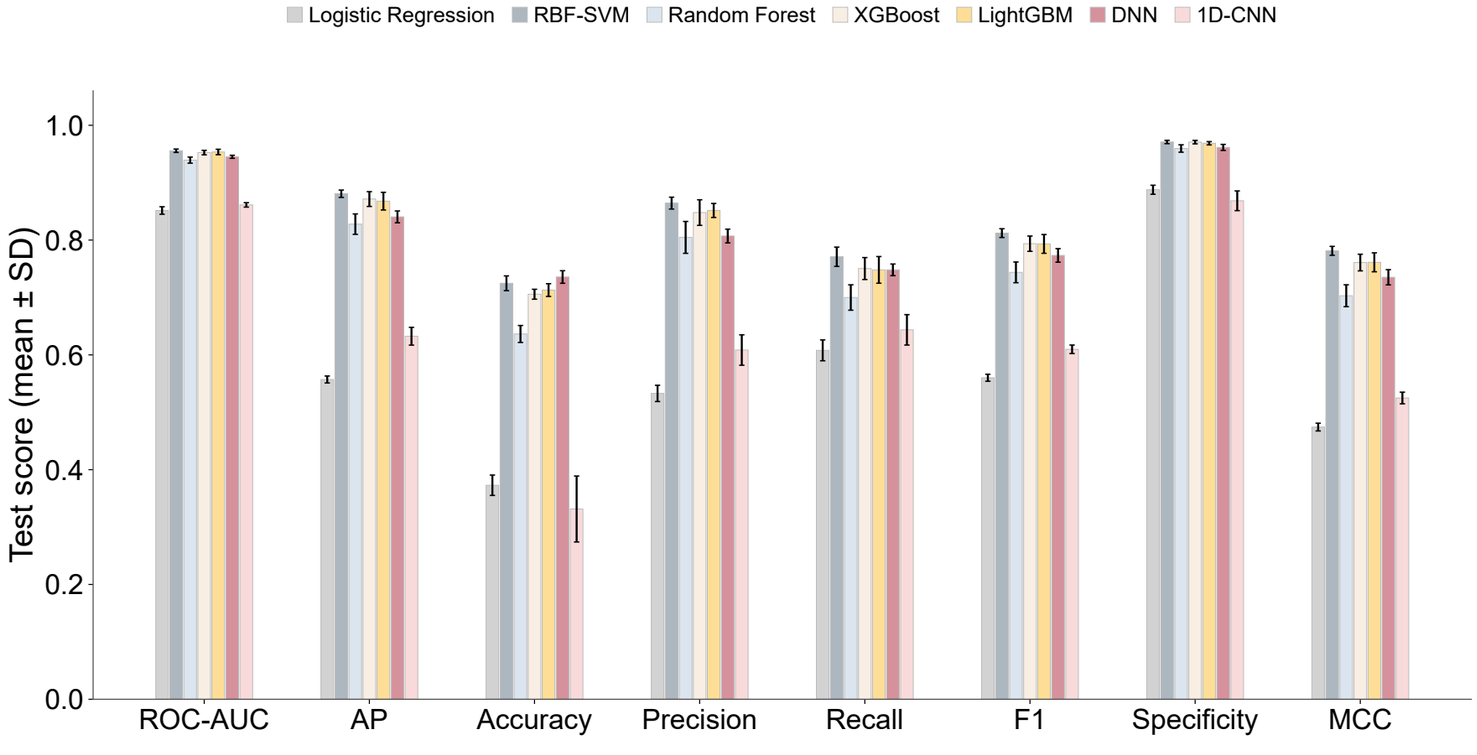
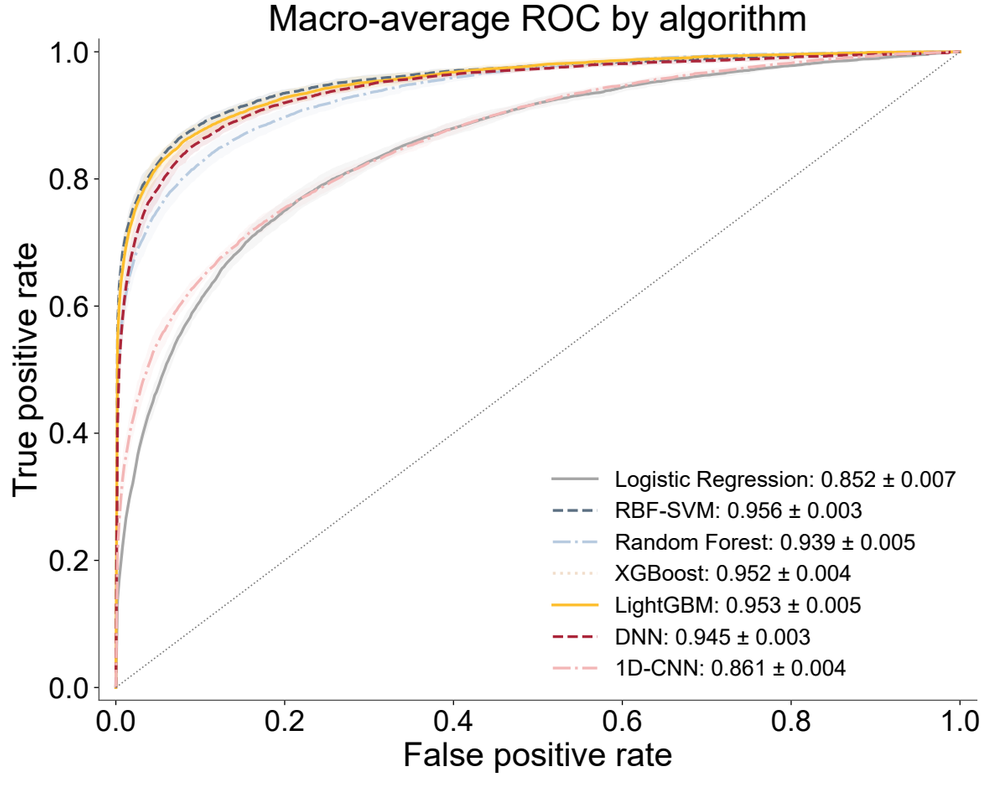
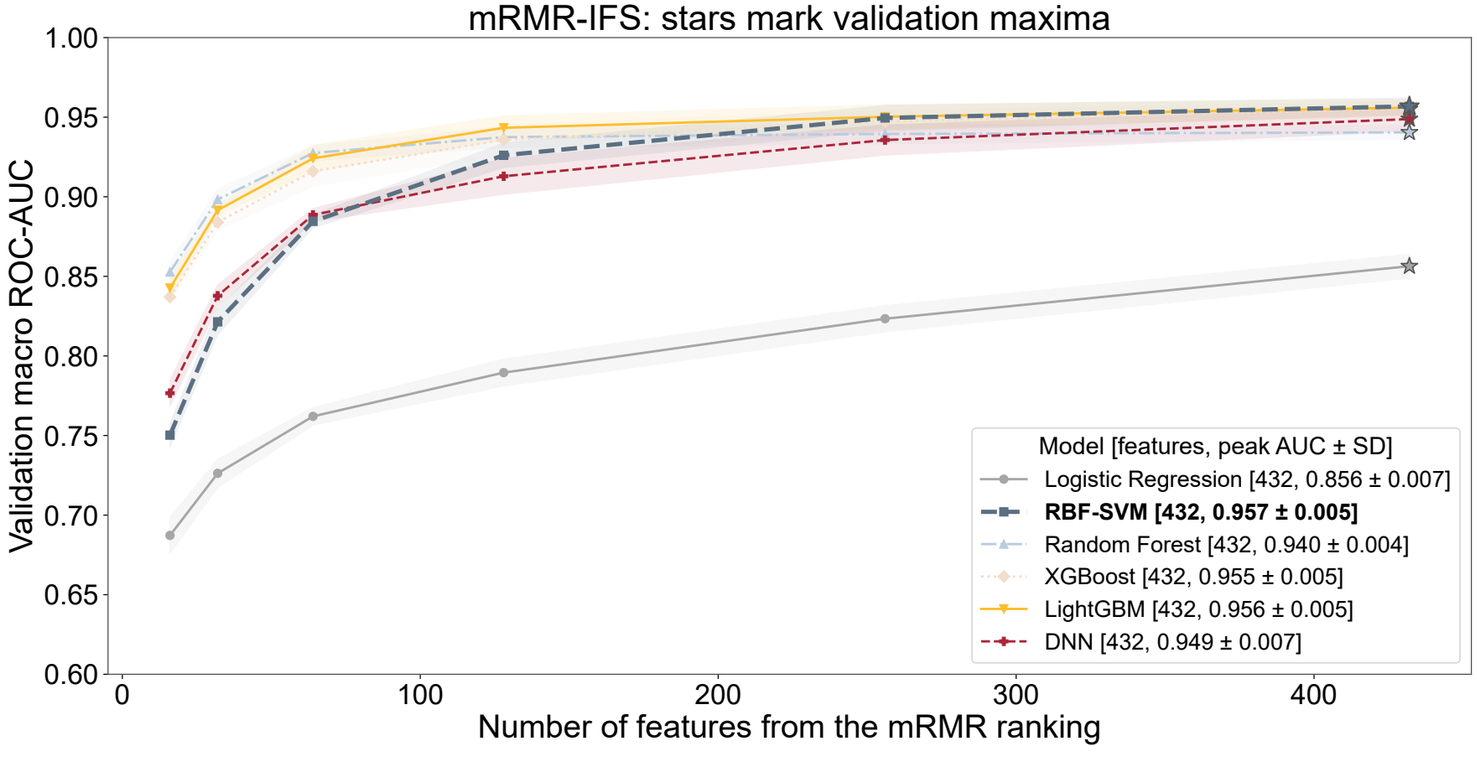
A


Logistic Regression
RBF-SVM
Random Forest
XGBoost
LightGBM
DNN
1D-CNN
Test score (mean ± SD)
1.0
0.8
0.6
0.4
0.2
0.0
ROC-AUC
AP
Accuracy
Precision
Recall
F1
Specificity
MCC

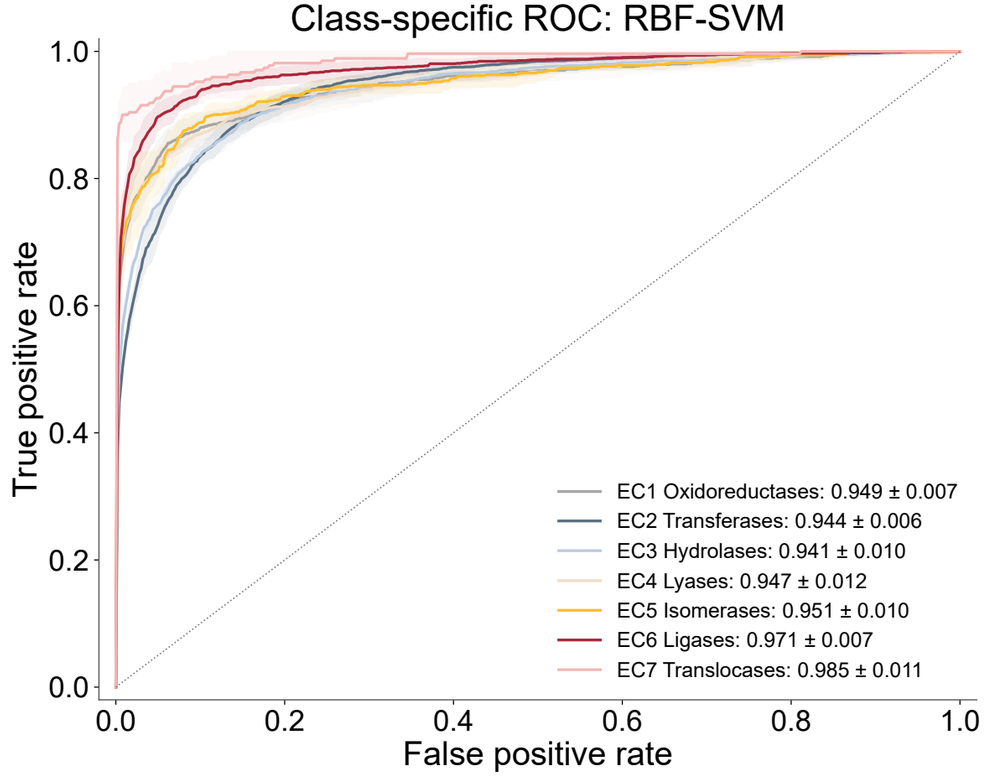
B


Macro-average ROC by algorithm
True positive rate
False positive rate
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression: 0.852 ± 0.007
RBF-SVM: 0.956 ± 0.003
Random Forest: 0.939 ± 0.005
XGBoost: 0.952 ± 0.004
LightGBM: 0.953 ± 0.005
DNN: 0.945 ± 0.003
1D-CNN: 0.861 ± 0.004

C


mRMR-IFS: stars mark validation maxima
Validation macro ROC-AUC
Number of features from the mRMR ranking
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0
100
200
300
400
Model [features, peak AUC ± SD]
Logistic Regression [432, 0.856 ± 0.007]
RBF-SVM [432, 0.957 ± 0.005]
Random Forest [432, 0.940 ± 0.004]
XGBoost [432, 0.955 ± 0.005]
LightGBM [432, 0.956 ± 0.005]
DNN [432, 0.949 ± 0.007]

D


Class-specific ROC: RBF-SVM
True positive rate
False positive rate
1.0
0.8
0.6
0.4
0.2
0.0
EC1 Oxidoreductases: 0.949 ± 0.007
EC2 Transferases: 0.944 ± 0.006
EC3 Hydrolases: 0.941 ± 0.010
EC4 Lyases: 0.947 ± 0.012
EC5 Isomerases: 0.951 ± 0.010
EC6 Ligases: 0.971 ± 0.007
EC7 Translocases: 0.985 ± 0.011